Applications of systems of equations

Balancing chemical equations

Example 1. Propane is a common gas used for cooking and home heating. Each molecule of propane is comprised of 3 atoms of carbon and 8 atoms of hydrogen, written as C_3H_8. When propane burns, it combines with oxygen gas, O_2, to form carbon dioxide, CO_2, and water, H_2O. Balance the chemical equation

$$C_3H_8 + O_2 \to CO_2 + H_2O$$

that describes this process.

Solution: We need to find whole numbers a, b, c and d, so that the equation

$$aC_3H_8 + bO_2 \to cCO_2 + dH_2O$$

is balanced. Equating the number of carbon, hydrogen, and oxygen atoms on both sides of this equation yields the linear system

$$3a - c = 0, \qquad 8a - 2d = 0, \qquad 2b - 2c - d = 0$$

Solving this gives the solution set as

$$S = \left\{ t\left(\frac{1}{4}, \frac{5}{4}, \frac{3}{4}, 1\right) | t \in \mathbb{R} \right\}$$

Since whole numbers are required to balance the equation, we choose appropriate values of t: $t = 0, 4, 8, \cdots$. For example, if $t = 8$, then the balanced chemical equation is

$$2C_3H_8 + 10O_2 \to 6CO_2 + 8H_2O$$

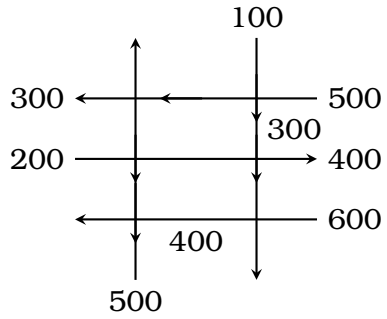
Traffic flow

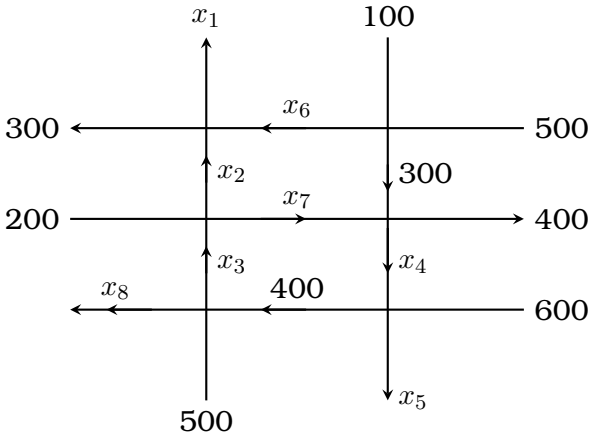
To study the flow of traffic through city streets, urban planners use mathematical models called directed graphs or digraphs. In these models, edges and points are used to represent streets and intersections, respectively. Arrows are used to indicate the direction of traffic. To balance a traffic network, we assume that

- outflow of each intersection is equal to the inflow,
- the total flow into the network is equal to the total flow out.

Example 2. Partial traffic flow information, given by average hourly volume, is known about a network of five streets, as shown in the figure. Complete the flow pattern for the network.

Solution: To complete the traffic model, we need to find values for the eight unknown flows, as shown below. Our assumptions about the intersections give us the





following set of linear equations

$$\begin{aligned} x_2 + x_6 &= 300 + x_1 \\ 100 + 500 &= x_6 + 300 \\ 200 + x_3 &= x_2 + x_7 \\ 300 + x_7 &= 400 + x_4 \\ 400 + 500 &= x_3 + x_8 \\ x_4 + 600 &= 400 + x_5 \end{aligned}$$

Moreover, balancing the total flow into the network with the total flow out gives us the additional equation

$$500 + 600 + 500 + 200 + 100 = 400 + x_5 + x_8 + 300 + x_1$$

The final linear system is

$$\begin{aligned} -x_1 + x_2 + x_6 &= 300 \\ x_6 &= 300 \\ x_2 - x_3 + x_7 &= 200 \\ -x_4 + x_7 &= 100 \\ x_3 + x_8 &= 900 \\ -x_4 + x_5 &= 200 \\ x_1 + x_5 + x_8 &= 1200 \end{aligned}$$

The solution is given by $x_1 = 1100 - s - t$, $x_2 = 1100 - s - t$, $x_3 = 900 - t$, $x_4 = -100 + s$, $x_5 = 100 + s$, $x_6 = 300$, $x_7 = s$ and $x_8 = t$

Notice that x_7 and x_8 are free variables. However, to obtain particular solutions, we must choose numbers for s and t that produce positive values for each x_i in the system (otherwise we will have traffic going in the wrong direction!) For example, $s = 400$ and $t = 300$ give a viable solution.

Electric circuits

In a direct current circuit containing resistances and sources of electromo- tive force (abbreviated EMF) such as batteries, a point at which three or more conductors are joined is called a node or branch point of the circuit, and a closed conduction path is called a loop. Any part of a circuit between two adjoining nodes is called a branch of the circuit.

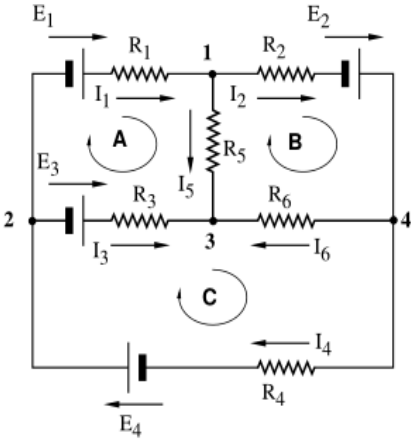
The circuit shown in Figure 2.6.1 is a typical example that contains four nodes, seven loops, and six branches.

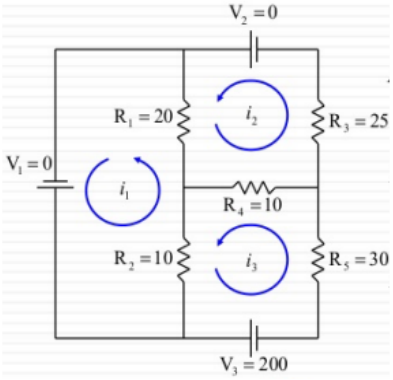
The problem is to relate the currents I_k in each branch to the resistances R_k and the EMFs E_k. This is accomplished by using **Ohm's law** and **Kirchhoff's rules** to produce a system of linear equations.

- **Ohm's Law:** Ohm's law states that for a current of I amps, the voltage drop (in volts) across a resistance of R ohms is given by $V = IR$.
- **Kirchhoff's rules**
  - **Node rule:** The algebraic sum of currents toward each node is zero. That is, the total incoming current must equal the total outgoing current. This is simply a statement of conservation of charge.
  - **Loop rule:** The algebraic sum of the EMFs around each loop must equal the algebraic sum of the IR products in the same loop. That is, assuming internal source resistances have been accounted for, the algebraic sum of the voltage drops over the sources equals the algebraic sum of the voltage drops over the resistances in each loop. This is a statement of conservation of energy.

We first fix a direction which will be called positive. If the positive direction is considered to be clockwise in each case, then applying the loop rule to the three indicated loops A, B, and C in the circuit shown above produces the three non-homogeneous equations in six unknowns: the I_k's are treated as the unknowns, while the R_k's and E_k's are assumed to be known.

$$\text{Loop A: } I_1R_1 - I_3R_3 + I_5R_5 = E_1 - E_3$$
$$\text{Loop B: } I_2R_2 - I_5R_5 + I_6R_6 = E_2$$
$$\text{Loop C: } I_3R_3 + I_4R_4 - I_6R_6 = E_3 + E_4$$





Example 3. Consider the following electric circuit We fix the *counterclockwise* direction as the positive direction. Note that the currents with sign associated with the resistance are as follows $R_1 : i_1, -i_2$; $R_2 : i_1, -i_3$; $R_3 : i_2$; $R_4 : i_2, i_3$ and $R_5 : i_3$. The resistance associated with the voltage are $V_1 : R_1, R_2$; $V_2 : R_1, R_3, R_4$ and $V_3 : R_2, R_4, R_5$. The voltage associated with the loops are left loop: V_1; upper right loop: V_2 and lower right loop: V_3. By Kirchoff's loop law we have the following system of equations

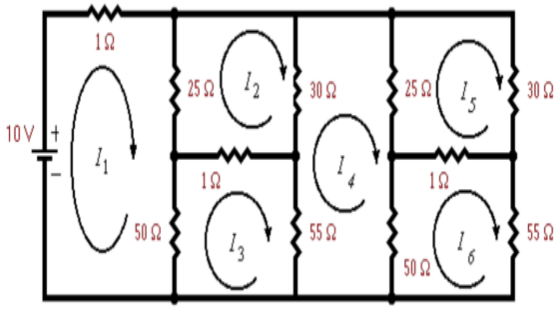
- For the left loop: $20(i_1 - i_2) + 10(i_1 - i_3) = 0$
- For the upper right loop: $25i_2 + 10(i_2 - i_3) + 20(i_2 - i_1) = 0$
- For the bottom right loop: $30i_3 + 10(i_3 - i_2) + 10(i_3 - i_1) = 200$

This gives us the system of equations

$$\begin{aligned} 30i_1 - 20i_2 - 10i_3 &= 0 \\ -20i_1 + 55i_2 - 10i_3 &= 0 \\ -10i_1 - 10i_2 + 50i_3 &= 0 \end{aligned}$$

which we solve and obtain the solution as $i_1 = 3$, $i_2 = 2$ and $i_3 = 5$.

Example 4. Consider the following circuit The positive direction is counter-clockwise.



Applying Kirchhoff's loop law, we get the following equations

$$\begin{aligned}
i_1 + 25(i_1 - i_2) + 50(i_1 - i_3) &= 10 \\
25(i_2 - i_1) + 30(i_2 - i_4) + 1(i_2 - i_3) &= 0 \\
50(i_3 - i_1) + 1(i_3 - 1_2) + 55(i_3 - i_4) &= 0 \\
55(i_4 - i_3) + 30(i_4 - 1_2) + 25(i_4 - i_5) + 50(i_4 - i_6) &= 0 \\
25(i_5 - i_4) + 30i_5 + 1(i_5 - i_6) &= 0 \\
50(i_6 - i_4) + 1(i_6 - i_5) + 55i_6 &= 0
\end{aligned}$$

Rearrange to obtain a system of equations and solve.

0.1 Cryptography: Hill Cipher

The Hill cipher is one of the classical cipher that is based on linear algebra. It involves the use of invertible matrices and modular arithmetic.

Modular arithmetic

Given an integer $n \geq 1$, called a modulus, two integers a and b are said to be *congruent modulo* m if their difference $a - b$ is divisible by m, that is there exists an integer k such that $a - b = km$. This is denoted by $a \equiv b \pmod{m}$ (read as a is congruent to b modulo m). For example $25 \equiv 5 \pmod{4}$ as 4 divides $25 - 5 = 20$ and $12 \equiv -3 \pmod{3}$ as 3 divides $12 - (-3) = 15$.

Since $a - b = km$, the congruence modulo relation $a \equiv b \pmod{m}$ can also written as $a = b + km$, which is known as the **division algorithm**. The formal definition is as follows.

The division algorithm

Definition 1. Let a, b be integers with $b > 0$. Then there exists **unique** integers q and r such that

$$a = bq + r, \quad 0 \leq r < b.$$

We note the following two concepts that will be used.

- Let $b > 0$. We say a modulo b is r if r is the remainder that is obtained when a is divided by b, that is there exists an integer q such that $a = bq + r$. For example, 52 modulo 5 is 2 as $52 = 5 \times 10 + 2$. If a modulo b is zero, then b divides a, that is a is a multiple of b. For example 52 modulo 26 is 0.

- The **modular multiplicative inverse** of an integer a is an integer x such that $ax \equiv 1 \pmod{m}$, that is m divides $ax-1$. For example, the modular multiplicative inverse of
  - 9 with respect to the modulus 26 is 3 as $9 \times 3 - 1 = 26$ is divisible by 26.
  - 8 with respect to the modulus 5 is 2 as $8 \times 2 - 1 = 15$ is divisible by 5.
  - 7 with respect to the modulus 12 is 7 as $7 \times 7 - 1 = 28$ is divisible by 12.

- A matrix K modulo m is the matrix obtained when each entry in K is replaced by the entry modulo m (recall this is the remainder r when the entry is divided by m and $0 \leq r < m$ by the division algorithm. For example if $m = 3$ or $m = 5$, then

$$\begin{pmatrix} 8 & 23 \\ 27 & 17 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 0 & 2 \end{pmatrix} \pmod{3}$$

$$\begin{pmatrix} 24 & 23 & 89 \\ 27 & 57 & 11 \\ 10 & 68 & 87 \end{pmatrix} = \begin{pmatrix} 4 & 3 & 4 \\ 2 & 2 & 1 \\ 0 & 3 & 2 \end{pmatrix} \pmod{5}$$

  The **inverse of a matrix modulo** m is as follows. Suppose $m = 26$ and the matrix is

$$K = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$$

  Then K^{-1} modulo 26 is found out as follows. First write the matrix inverse such that all the entries inside the matrix inverse are integers.

$$K^{-1} = \frac{1}{9} \begin{pmatrix} 5 & -3 \\ -2 & 3 \end{pmatrix}$$

  Replace $1/9$ by the modular multiplicative inverse of 9 with respect to 26 which is 3 ($9 \times 3 - 1$ is divisible by 26). Next, replace each entry of K^{-1} by the entry modulo 26 (the remainder r when the entry is divided by 26 with $0 \leq r > 26$ by the division algorithm). Thus
  - 5 modulo 26 is 5 as $5 = 26 \times 0 + 5$.
  - -3 modulo 26 is 23 as $-3 = 26 \times -1 + 23$.
  - -2 modulo 26 is 24 as $-2 = 26 \times -1 + 24$.
  - 3 modulo 26 is 3 as $3 = 26 \times 0 + 3$.

Thus,

$$K^{-1} \pmod{26} = 3 \begin{pmatrix} 5 & 23 \\ 24 & 3 \end{pmatrix} = \begin{pmatrix} 15 & 69 \\ 72 & 9 \end{pmatrix} = \begin{pmatrix} 15 & 17 \\ 20 & 9 \end{pmatrix} \pmod{26}$$

With this background, we discuss examples on Hill cipher. It involves the following two information that is known to both the sender and receiver.

- Assigning the $A = 0$, $B = 1$, $\cdots$, $Z = 25$.
- A key matrix K modulo 26 which is also invertible modulo 26.

Example 5. Carry out encryption and decryption by Hill cipher for the message **HELP** using the key matrix

$$K = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$$

The steps are as follows.

- First, write the given matrix modulo 26. In the present case $K \pmod{26} = K$ as each entry of K modulo 26 is the entry itself.
- Since the key matrix has two columns, we will write the message in numerical values as a matrix M with 2 rows (to make the matrices K and M compatible for the product KM).

$$M = \begin{pmatrix} H & L \\ E & P \end{pmatrix} = \begin{pmatrix} 7 & 11 \\ 4 & 15 \end{pmatrix}$$

- Multiply the key matrix (modulo 26) on the left for encryption. The matrix product is also written modulo 26.

$$KM = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 7 & 11 \\ 4 & 15 \end{pmatrix} = \begin{pmatrix} 33 & 78 \\ 34 & 97 \end{pmatrix} = \begin{pmatrix} 7 & 0 \\ 8 & 19 \end{pmatrix} \pmod{26} = \mathfrak{E}$$

- Get the letter corresponding to the numbers in the encrypted matrix:

$$7 \rightsquigarrow H, \quad 8 \rightsquigarrow I, \quad 0 \rightsquigarrow A, \quad 19 \rightsquigarrow T.$$

  The encrypted message that will be sent is **HIAT**.

- The receiver receives the encrypted message $\mathfrak{E}$ as the matrix product $KM \pmod{26}$ and decodes it as

$$\mathfrak{E} = KM \pmod{26} \Rightarrow M = K^{-1} \pmod{26} \mathfrak{E}$$

  The receiver writes the encrypted message as a matrix with two rows (to make the matrix product $K^{-1} \pmod{26} \mathfrak{E}$ compatible) and the corresponding numerical values. Thus

$$\begin{pmatrix} H & A \\ I & T \end{pmatrix} \rightsquigarrow \begin{pmatrix} 7 & 0 \\ 8 & 19 \end{pmatrix}$$

- The matrix inverse K^{-1} has been found earlier as

$$K^{-1} = \begin{pmatrix} 15 & 17 \\ 20 & 9 \end{pmatrix} (\mod 26)$$

  The decryption is done as

$$M = \begin{pmatrix} 15 & 17 \\ 20 & 9 \end{pmatrix} \begin{pmatrix} 7 & 0 \\ 8 & 19 \end{pmatrix} = \begin{pmatrix} 241 & 323 \\ 212 & 171 \end{pmatrix} = \begin{pmatrix} 7 & 11 \\ 4 & 15 \end{pmatrix} (\mod 26)$$

  The receiver recovers the original message as **HELP**.

Example 6. Carry out the encryption and decryption by Hill cipher for the message **ACT** with the key matrix

$$K = \begin{pmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{pmatrix}$$

The steps are briefly outlined below.

- It is checked that the key matrix K modulo 26 is the matrix K itself.
- The message in numerical values is

$$M = \begin{pmatrix} A \\ C \\ T \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 19 \end{pmatrix}$$

- The encrypted matrix $\mathfrak{E} = KM$ is

$$\mathfrak{E} = KM = \begin{pmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \\ 19 \end{pmatrix} = \begin{pmatrix} 67 \\ 222 \\ 319 \end{pmatrix} = \begin{pmatrix} 15 \\ 14 \\ 7 \end{pmatrix} (\mod 26)$$

- The encrypted message sent is **POH**.
- The matrix inverse modulo 26 of K is given by

$$K^{-1} = \begin{pmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{pmatrix} (\mod 26)$$

- The decryption is carried out as

$$M = \begin{pmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{pmatrix} \begin{pmatrix} 15 \\ 14 \\ 7 \end{pmatrix} = \begin{pmatrix} 260 \\ 574 \\ 539 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 9 \end{pmatrix} (\mod 26)$$

  and the original message is recovered as ACT.

Example 7. Encrypt and decrypt the message **TIME TO STUDY** using the Hill cipher key

$$K = \begin{pmatrix} 1 & 4 & 0 \\ 7 & 11 & 2 \\ 0 & 5 & 1 \end{pmatrix}$$

The steps are briefly outlined below.

- Read the message without space as **TIMETOSTUDY** and write the matrix with numerical values.

$$M = \begin{pmatrix} 19 & 4 & 18 & 3 \\ 8 & 19 & 19 & 24 \\ 12 & 14 & 20 & ? \end{pmatrix} \rightsquigarrow M = \begin{pmatrix} 19 & 4 & 18 & 3 \\ 8 & 19 & 19 & 24 \\ 12 & 14 & 20 & 23 \end{pmatrix}$$

  For the matrix product KM to be defined, M must have 3 rows. The $?$ appears because the message ends before M can be written as a 3×4 matrix. It is customary to fill such spaces with 23 corresponding to the letter X.

- **Encryption:** $\mathfrak{E} = KM(\mod 26)$ is

$$\begin{pmatrix} 1 & 4 & 0 \\ 7 & 11 & 2 \\ 0 & 5 & 1 \end{pmatrix} \begin{pmatrix} 19 & 4 & 18 & 3 \\ 8 & 19 & 19 & 24 \\ 12 & 14 & 20 & 23 \end{pmatrix} = \begin{pmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{pmatrix} (\mod 26)$$

- **Decryption:**

$$K^{-1} = \frac{1}{27} \begin{pmatrix} -1 & 4 & -8 \\ 7 & -1 & 2 \\ -35 & 5 & 17 \end{pmatrix} = \begin{pmatrix} 25 & 4 & 18 \\ 7 & 25 & 2 \\ 17 & 5 & 17 \end{pmatrix} (\mod 26)$$